Improving the Upper Bound on the 5×5 Complete-Pivoting Growth Factor via LP Certificates and Branch-and-Bound

Hans Sihan Zhu*

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Abstract

The growth factor of Gaussian elimination with complete pivoting measures how large the entries can become during elimination, and its worst case g_n is known exactly only for $n \leq 4$. For $n = 5$, Chen, Edelman and Urschel recently showed $4.1325\ldots \leq g_5 \leq 4.84$, the lower bound coming from an explicit matrix that they conjecture to be optimal. We prove $g_5 \leq 4.15$, reducing the gap between the known bounds from about 0.71 to 0.018. After a reduction of completely pivoted 5×5 matrices to a normal form described by 24 parameters of their LU factorization, we relax the bilinear constraints on boxes of parameters to linear programs, and exhaust the parameter space by branch-and-bound. Each pruned box and each tightening is justified by an exact certificate, a vector of nonnegative multipliers, whose validity follows from LP weak duality. A floating-point LP solver proposes the certificates but is never trusted, and the results are considered rigorously proven only through the independent verifier, which verifies the prunings and tightenings using exact rational arithmetic. The process takes 3.5 million certificates for $K = 4.15$ and roughly 10^5 for $K = 4.5$. The proof uses only the classical values $g_3 = \frac{9}{4}$, $g_4 = 4$ and an inequality of Cohen.

1 Introduction

Pivoting is a technique used to minimize growth during Gaussian elimination. For a nonsingular matrix $A \in \mathbb{R}^{n\times n}$, let elimination produce the Schur complements $A^{(k)}$ for $k = 1, \cdots, n$, where $A^{(k)}$ is the matrix after finishing the subtraction of $k-1$ pivots, with only indices $[k, n] \times [k, n]$; specially, $A^{(1)} = A$.

With the purpose of minimizing error in elimination, complete pivoting is the process of permuting the rows and columns $k, \cdots, n$ so that

$$|a_{kk}^{(k)}| = \max_{i,j\geq k} |a_{ij}^{(k)}|,$$

where any maximizer may be chosen when there are ties. The pivots are $p_k = |a_{kk}^{(k)}| \geq 0$ The growth factor and its extremal value are

$$\rho(A) = \frac{\max_k \max_{i,j} |a_{ij}^{(k)}|}{\max_{i,j} |a_{ij}|}, \qquad g_n = \sup_{A,\text{legal complete pivot sequences}} \rho(A).$$

Complete pivoting produces the factorization $LU = PAQ$, where P, Q are permutation matrices, L lower triangular and U upper triangular. It is easy to see that $g_2 = 2$, and Cohen 1974 unifies $g_3 = \frac{9}{4}$, $g_4 = 4$ and $g_5 < 5.005$, including results from Tornheim and Cryer [2, 3, 4]. $n = 5$ is the smallest positive integer whose maximal growth has not yet been determined. In 1969, Tornheim's report gives an upper bound of $4\frac{17}{18} = 4.94$ [5]. Recently, Chen, Edelman and

*Shanghai High School International Division, hanssihanzhu@gmail.com

Urschel (2026) conjecture that $\gamma \approx 4.132517$, which is the root of a 61st degree polynomial, may be the actual value of g_5; they also show the result $g_5 \leq 4.84$ [1].

In this paper, we propose a method that could at least produce the bound $g_5 \leq 4.15$. Although also produced for 4.2 and 4.3, only certificates for 4.5 and 4.15 are exported, and the bounds are independently verified for 4.5 and 4.15.

2 Reducing the Problem to a 24-dimensional Domain

We normalize any 5×5 matrix and choices of pivots (considering ties) into a more workable matrix that would yield the same growth factor without pivoting. Formally:

Lemma 2.1. *For every nonsingular A and every legal complete-pivoting sequence S there exists a matrix A' with sequence S' with the same ρ that needs no pivoting and satisfies:*

1. $|A'^{(k)}_{i,j}| \leq A'^{(k)}_{kk}$, *where $A'^{(k)}_{kk}$ is its pivot, for all $i, j \geq k$. We will call this the CP condition.*
2. $A'_{11} = 1$
3. *All pivots are positive, and $A'_{i1} \geq 0$ for all i.*

Proof. Fix a complete-pivoting run on A. Label each original row and column by the step at which it becomes a row or column. Arrange them in that order in the beginning to have $B = PAQ$. One can actually show that these P, Q are exactly the same as the ones we were working with in the introduction.

B can run complete pivoting without pivoting, as the entries along its diagonal will always correspond to the exact same pivot rows and columns as in the original run. Furthermore, elimination with those pivots updates each remaining entry using only that entry, its pivot-column entry and its pivot-row entry, yielding identical entries at each step:

$$b_{ij} \to b_{ij} - \frac{b_{ik} b_{kj}}{b_{kk}}$$

Hence $\rho(A, S) = \rho(B, S')$ where S' is always choosing diagonal entries as pivots, because each run produces the exact same remaining entries.

Furthermore, the same ρ-is-same argument can be repeated with the transformation of multiplying a row or column by -1. Suppose that we transform

$$x_{ij} = r_i x_{ij} c_j$$

with $r_i, c_i \in \{-1, 1\}$, the individual subtractions in elimination then become

$$x'_{ij} = r_i c_j x_{ij} - \frac{r_i c_k x_{ik} r_k c_j x_{kj}}{r_k c_k x_{kk}} = r_i c_j \left(x_{ij} - \frac{x_{ik} x_{kj}}{x_{kk}} \right)$$

Since there are only eliminations in B, using $D = RBC$ where R is the row sign diagonal matrix and C is the column sign diagonal matrix means that all operations effectively only perform on B, and since R and C are only sign matrices, no intermediate magnitudes are affected and all properties are preserved. Hence $\rho(B, S') = \rho(D, S')$. We choose such a D so that the first column and the pivots are positive: choose R such that the first column is all positive, then C such that the 2nd to nth pivots are positive.

The only case that gives residual freedom is when B_{i1} is 0, but this does not affect any of the important properties as long as the horizontal and vertical multipliers are properly matched.

Finally, we divide with $A' = \frac{1}{|D_{11}|} D$, which just normalizes the entries so that $A'_{11} = 1$. ρ is clearly unchanged.

We have $\rho(A, S) = \rho(A', S')$, where (A', S') satisfies all desired properties in the Lemma. We will use A to implicitly include the assumption that S is all diagonal entries henceforth.

□

With this normalization, we define the domain $\mathcal{F}$ as follows. Let $A \in \mathbb{R}^{5\times 5}$ be in the normal form of Lemma 2.1. Let $A = LU$ be its elimination factorization and define $y = (\ell, p, u) \in \mathbb{R}^{10} \times \mathbb{R}^4 \times \mathbb{R}^{10} = \mathbb{R}^{24}$ by

$$\ell_{it} = \frac{a_{it}^{(t)}}{a_{tt}^{(t)}} \ (1 \le t < i \le 5), \qquad p_t = a_{tt}^{(t)} \ (t = 2, \cdots, 5), \qquad u_{tj} = a_{tj}^{(t)} (1 \le t < j \le 5),$$

with $p_1 = u_{11} = a_{11} = 1$ already known, and with the conventions of $\ell_{ii} = 1$, $u_{tt} = p_t$, $p_1 = u_{11} = 1$ (these variables are excluded from the definition of y but would prove highly useful for later notation).

Under this definition, we prove a few important properties that provide restrictions on y for us to find a stronger enclosing domain.

Lemma 2.2.

$$a_{ij}^{(k)} = \sum_{t=k}^{\min(i,j)} \ell_{it} u_{tj} \qquad (k \le i, j).$$

Proof. We consider

$$a_{i,j}^{(k+1)} = a_{i,j}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kj}^{(k)} = a_{i,j}^{(k)} - \ell_{i,k} u_{k,j}.$$

Rewriting and telescoping until $m = \min(i, j)$, there is

$$a_{i,j}^{(k)} = a_{i,j}^{(m)} + \ell_{i,m-1} u_{m-1,j} + \cdots + \ell_{i,k} u_{k,j}.$$

If $m = i$, $a_{i,j}^{(m)} = u_{i,j} = u_{m,j}$ and $\ell_{i,m} = \ell_{i,i} = 1$. If $m = j$, $a_{i,j}^{(m)} = a_{i,j}^{(j)} = \frac{a_{ij}^{(j)}}{a_{jj}^{(j)}} a_{jj}^{(j)} = \ell_{i,j} u_{j,j} = \ell_{i,m} u_{m,j}$. In both cases, there is

$$a_{i,j}^{(k)} = \ell_{i,m} u_{m,j} + \ell_{i,m-1} u_{m-1,j} + \cdots + \ell_{i,k} u_{k,j},$$

as desired.

□

Corollary 2.3. *CP implies that $|\ell_{ik}| \le 1$ for $i > k$ and $|u_{kj}| \le u_{kk} = p_k$ for $j > k$.*

Proof. Since

$$\ell_{i,k} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$$

and previous permutations guarantee that $a_{kk}^{(k)}$ is the value with largest magnitude in $a_{[k,n]\times[k,n]}^{(k)}$, there is

$$|\ell_{i,k}| = \frac{|a_{ik}^{(k)}|}{|a_{kk}^{(k)}|} \le 1.$$

Also, $u_{kj} = a_{kj}^{(k)}$ and $u_{kk} = a_{kk}^{(k)}$. Again by $a_{kk}^{(k)}$'s maximality in the remaining matrix $A^{(k)}$, there is $|u_{k,j}| \le |u_{kk}| = p_k$. □

We define $\mathcal{F}$ as the set of all $y(A) \in \mathbb{R}^{24}$ for all valid normalized CP matrices A.

Corollary 2.4. *For all $y \in \mathcal{F}$, there is*

$$\ell_{i1} \geq 0, \qquad |\ell_{it}| \leq 1, \qquad p_t > 0, \qquad \left| \sum_{t=k}^{\min(i,j)} \ell_{it} u_{tj} \right| \leq p_k \ (1 \leq k \leq 4, k \leq i, j \leq 5)$$

Proof. We show that all constraints are satisfied due to the properties of normalized A:

1. $p_t > 0$. $p_t = a_{tt}^{(t)}$ is positive in normalized A (Lemma 2.1).
2. $\ell_{i1} \geq 0$. $\ell_{i1} = \frac{a_{i1}^{(1)}}{a_{11}^{(1)}} = a_{i1}^{(1)}$ since $a_{11}^{(1)} = 1$, and normal form guarantees $a_{i1} \geq 0$.
3. $|\ell_{it}| \leq 1$ is true by (Corollary 2.3)
4. Lemma 2.2 gave us that $\left|\sum_{t=k}^{\min(i,j)} \ell_{it} u_{tj}\right| = a_{ij}^{(k)} \leq a_{kk}^{(k)} = p_k$.

□

This means that every normalized A has its components $y = (\ell, u, p) \in \mathbb{R}^{24}$ satisfying the above properties, implying that

$$g_5 = \sup_A \max_k p_k = \sup_{y \in \mathcal{F}} \max_k p_k \tag{1}$$

where $\rho(A) = \max_k p_k$ because p_k is the maximal entry of $A_{ij}^{(k)}$. If we wish to show that $g_5 \leq K$ for some upper bound K, then we simply show that $\sup_{y \in \mathcal{F}} \max_k p_k \leq K$. This follows from showing that there does not exist a y such that $\max p_k \geq K$.

Furthermore, there is $p_2, p_3, p_4 \leq g_4 = 4$ because the leading 4×4 sub-matrix of A with CP has CP because its Schur complements are the leading blocks of $A^{(k)}$; the larger CP is a stronger version of the smaller CP. With CP, we can apply $\rho(A) = \max(p_1, p_2, p_3, p_4)$ and we have that $g_4 = 4 \geq \rho(A)$.

Yet, since $K > \max(p_1, p_2, p_3, p_4)$, for $\max p_k \geq K$, there must be $p_5 \geq K$. The following is then sufficient to show that $g_5 \leq K$:

Proposition 2.5. *For given $K \in \{4.5, 4.15\}$ the set $\mathcal{F}_K := \{ y \in \mathcal{F} : p_5 \geq K \}$ is empty.*

3 Linear Relaxation within Boxes

We characterize regions containing y using "boxes". A box B can be represented as:

$$B = \prod_{i=1}^{24} [a_i, b_i] = \{y \in \mathbb{R}^{24} : a_i \leq y_i \leq b_i \text{ for all } i = 1, \ldots, 24\}.$$

Given a box B, we extend the box to 60 dimensions by applying the R6 rows (as defined later), then find a series of linear constraints on the 60 dimensions (R1-R5) to apply Linear Programming.

Formally, let

$$P = \{(i, t, j) : 1 \leq t < i \leq 5,\ t \leq j \leq 5,\ (t, j) \neq (1, 1)\},$$

and introduce a variable w_{itj} for each $(i, t, j) \in P$, standing for $\ell_{it} u_{tj}$, this product divided by the kth pivot represents precisely how much was subtracted from entry $u_{ij}^{(t)}$.

We use Φ to represent the expansion from y to x, namely

$$x = \Phi(y) = (y, w),$$

and the LP variables are then $x = (\ell, p, u, w) \in \mathbb{R}^{60}$. For convenience, set

$$\tau_{itj} = \begin{cases} u_{ij}, & \text{if } t = i, \\ w_{itj}, & \text{otherwise,} \end{cases}$$

as w was not defined for $t = i$.

We now derive six groups of constraints (R1-R6) that must hold for any $y \in \mathcal{F}_K \cap B$, together giving 72 linear expansion rows (R6) and 328 linear inequalities (R1-R5) in the 60 variables of x; any information from the box B is considered constant. These rows are collectively referred to as the relaxation of a box:

$$R(B) = \{x \in \mathbb{R}^{60} : \underline{x} \le x \le \overline{x}, Gx \le h\}.$$

In particular, R6 completes B of 24 dimensions into B^* of 60 dimensions, and these make up the $\underline{x}$ and $\overline{x}$. The rest of the rows constitute G.

3.1 R1: CP Rows

By the CP property and *Lemma* 2.2, we have

$$\pm \sum_{t=k}^{\min(i,j)} \tau_{itj} \le \left| \sum_{t=k}^{\min(i,j)} \tau_{itj} \right| = |a_{ij}^{(k)}| \le p_k, \qquad 1 \le k \le 4,\ k \le i \le 5,\ k < j \le 5.$$

Importantly, the constraint is purely linear and involves only a subset of dimensions of x. This gives a total of $2 \times (20 + 12 + 6 + 2) = 80$ rows.

3.2 R2: Column Rows

For $t \ge 2$, where w is not trivially within magnitude 1, there is for $i > t$

$$\pm w_{itt} \le |w_{itt}| = |\ell_{it} u_{tt}| = |\ell_{it} p_t| \le p_t,$$

giving a total of $2(3 + 2 + 1) = 12$ rows.

3.3 R3: Pivot Rows

In this subsection, we impose constraints between the pivot variables p_i, taking inequalities from literature while deriving others. We first present two lemmas, then systematize into rows, as with the rows before.

Lemma 3.1. *Let $k \ge 1$ and $k + m - 1 \le 5$. Let W be the leading $m \times m$ block of $A^{(k)}$, divided by p_k. Then W is the normalized CP matrix with pivots $1, \frac{p_{k+1}}{p_k}, \cdots, \frac{p_{k+m-1}}{p_k}$. Hence*

$$p_{k+m-1} \le g_m p_k.$$

Proof. All elimination is scaled by $\frac{1}{p_k}$, it is easy to show (inductively) that all results will be scaled by that factor. The choice of pivots will not be affected, they will remain on the diagonal as guaranteed by the CP condition and the fact that scaling does not change relative magnitude. □

We briefly derive $p_2 \le 2$. Let normalized $A = \begin{pmatrix} 1 & a \\ b & ab + p_2 \end{pmatrix}$. Since $|ab + p_2| \le p_1 = 1$, $|p_2| \le 2$.

Indeed, $g_2 = 2$, and furthermore, $g_3 = \frac{9}{4}, g_4 = 4$. These imply that $p_3 \le \frac{9}{4}$ and $p_4 \le 4$.

Using Lemma 3.1, there is also

$$p_5 \le g_3 p_3 = \frac{9}{4} p_3 \le \frac{81}{16}$$

and

$$p_5 \le g_4 p_2 = 4 p_2.$$

Lemma 3.2. *(Cohen's inequality). Every normalized CP 3×3 matrix with pivots $1, q, s$ satisfies $s \le 3q - q^2$.*

Proof. From Cohen 1974 [2, Lemma 2.1 and eq. (11)], we already have $s \le 3q - q^2$ for $s \ge 2$. We prove the rest by considering the cases for $q \le 2$:

- If $q \le 1$, then $s \le 2q \le 3q - q^2$
- If $1 \le q \le 2$ and $s \le 2$, then $s \le 2 \le 3q - q^2$

□

For our rows, we consider $m = 2, 3, 4$.

Because $3q - q^2 = r^2 + (3 - 2r)q - (q - r)^2 \le r^2 + (3 - 2r)q$ from *Lemma* 3.2, s can be bounded above by a linear expression in q for some constant r. Applying this to windows ($k = 1, 2, 3, m = 3$) and $r \in S = \{\frac{1}{2}, \frac{3}{4}, 1, \frac{5}{4}, \frac{11}{8}, \frac{3}{2}, \frac{13}{8}, \frac{7}{4}, 2\}$, we obtain 27 rows describing the relationship between pivots two indices apart ($m = 3$ case).

When $m = 2$, there is $p_{k+1} \le 2p_k$ for $k = 1, 2, 3, 4$, and when $m = 4$, there is $p_5 \le 4p_2$.

This totals $27 + 4 + 1 = 32$ rows

3.4 R4: McCormick Rows

These rows are crucial to ensuring the relationship between w and ℓ, u, using the interval of two variables to bound their product, as originally proposed by McCormick [6]. Fix $(i, t, j) \in P$, let $\ell_{it} \in [a, b]$ and $u_{tj} \in [c, d]$ be the intervals in B. There is $w_{itj} = \ell_{it} u_{tj}$ per our definition (which we will for this section without indices), which gives

$$(\ell - a)(u - c) = w - au - c\ell + ac \ge 0 \tag{2}$$

$$w \ge c\ell + au - ac. \tag{3}$$

Analogously,

$$w \ge d\ell + bu - bd.$$

On the other hand, there is

$$(\ell - a)(u - d) = w - au - d\ell + ad \le 0 \tag{4}$$

$$w \le au + d\ell - ad \tag{5}$$

Analogously,

$$w \le c\ell + bu - bc.$$

Across all 36 possible P, this gives 144 rows of constraints.

3.5 R5: Perspective Rows

With the same setup in R4, except that we impose additionally that $j > t$ as when $j = t$, the R2 inequalities dominate. Let $m = \frac{a+b}{2}$ and $\delta = \frac{b-a}{2} \geq 0$, which are constants. Consider

$$w - mu = u(\ell - m),$$

which leads to

$$\pm(w - mu) \leq |w - mu| \leq |(\ell - m)u| \leq \delta p_t.$$

This gives $30 \times 2 = 60$ rows.

3.6 R6: Expansion Rows

Suppose that $\ell_{it} \in [a, b]$ and $u_{tj} \in [c, d]$, where the interval bounds are constants taken from B. Then,

$$w_{itj} \in [\min\{ac, ad, bc, bd\}, \max\{ac, ad, bc, bd\}].$$

This gives 36 pairs of rows, which we call the Expansion Rows because they effectively establish the intervals up to 60 dimensions. This family of rows is always computed from B to obtain the extended box B^*, which is reflected in the overall relaxation. We formalize this notion in Lemma 3.3.

Lemma 3.3. *If $y \in \mathcal{F}_K \cap B$, then $\Phi(y) \in R(B)$.*

Proof. Each subsection R1-R6 shows its rows hold at $\Phi(y)$. They only use $y \in B$ (for the box intervals in R4-R6) and $y \in \mathcal{F}$ (for CP and the pivot theorems in R1-R3).

□

4 LP Weak Duality Lemma

We present the core lemma that proves impossibility as desired in our proposition. The R represents precisely the relaxation that we have already defined for a box B.

Theorem 4.1. *Let $R = \{x : Gx \leq h,\ \underline{x} \leq x \leq \overline{x}\}$, $c \in \mathbb{R}^{60}$, $\lambda \geq 0$, and $d = c - G^\top \lambda$. Then*

$$\sup_{x \in R} c^\top x \leq \lambda^\top h + \sum_{v=1}^{60} \max(d_v \underline{x}_v,\ d_v \overline{x}_v).$$

Proof. Since $\lambda \geq 0$, for all $x \in R$, there is $Gx \leq h$, giving

$$\lambda^\top Gx \leq \lambda^\top h.$$

The definition of d lets us rearrange to obtain

$$d^\top x + \lambda^\top Gx = c^\top x.$$

Combining yields

$$c^\top x = d^\top x + \lambda^\top Gx \leq d^\top x + \lambda^\top h.$$

Considering all $x \in R$, there is

$$\sup_{x \in R} c^\top x \leq \lambda^\top x + \max\left(d^\top x\right) = \lambda^\top h + \sum_{v=1}^{60} \max(d_v \underline{x}_v,\ d_v \overline{x}_v). \tag{6}$$

□

In Theorem 4.1, let

$$Q(c, \lambda) = \lambda^\top h + \sum_{v=1}^{60} \max(d_v \underline{x}_v,\ d_v \overline{x}_v),$$

the RHS of Equation (6). Then an important implication is that when $c = 0$, then showing $Q(0, \lambda) < 0$ for some $\lambda \geq 0$ shows that $R(B) = \emptyset$. The λ obtained is what we call a "certificate", or just the dual variable of x in ordinary weak duality.

Although unrelated to correctness, Farkas' Lemma [8] [9] gives that when $R(B) = \varnothing$ then there exists $\lambda \geq 0$ with $Q(0, \lambda) < 0$, showing that in principle, there is always a valid certificate to be found.

5 The Pruning Algorithm

From our analysis in Section 3.3, there is $p_2 \geq K/4, p_3 \geq 4K/9, p_4 \geq K/2$. Let $\lfloor x \rfloor_*$ denote rounding down to the grid $2^{-30}\mathbb{Z}$, strictly below x. The root box $B_0(K) \subset \mathbb{R}^{24}$ then is:

$$\begin{aligned}
&\ell_{i1} \in [0, 1], \quad \ell_{it} \in [-1, 1]\ (t \geq 2),\\
&p_2 \in \left[\lfloor K/4 \rfloor_*,\, 2\right], \quad p_3 \in \left[\lfloor 4K/9 \rfloor_*,\, \tfrac{9}{4}\right], \quad p_4 \in \left[\lfloor K/2 \rfloor_*,\, 4\right], \quad p_5 \in \left[K,\, \tfrac{81}{16}\right],\\
&u_{1j} \in [-1, 1], \qquad u_{tj} \in [-\bar{p}_t,\ \bar{p}_t] \ \ (t \geq 2),
\end{aligned}$$

where $\bar{p}_t$ is the upper end of the interval of p_t. By Corollary 2.3 and pivot bounds above, $\mathcal{F}_K \subseteq B_0(K)$. For $K = \frac{9}{2}$, for instance:

$$p_2 \in [\tfrac{9}{8} - 2^{-30},\, 2], \quad p_3 \in [2 - 2^{-30},\, \tfrac{9}{4}], \quad p_4 \in [\tfrac{9}{4} - 2^{-30},\, 4], \quad p_5 \in [\tfrac{9}{2},\, \tfrac{81}{16}].$$

Notably, we take a strict round-down because of our computational tools, not because of the mathematics: IEEE-754 Float64s may have rounded up when finding the lower bound, leading to inaccuracies of up to $\frac{1}{2}$ulp.

Since $\mathcal{F}_K \subseteq B_0(K)$, showing that

$$\mathcal{F}_K \cap B_0(K) = \varnothing \tag{7}$$

means that

$$\mathcal{F}_K = \varnothing$$

as desired.

By Lemma 3.3 and Theorem 4.1, a box B contains no point of $\mathcal{F}_K$ if some $\lambda \geq 0$ gives $Q(0, \lambda) < 0$ or $Q(e_{p_5}, \lambda) < K$. We cover $B_0(K)$ by melding such smaller boxes, as will be formalized in Theorem 5.1.

5.1 HiGHS and rough certification

We will make use of the HiGHS program [7] to generate approximate certificates. The input is always one LP $(\underline{x}, \overline{x}, G, h, t)$, set up per box for 61 variables: the 60 coordinates of x plus the elastic number t. Bounds on the unknown are given by $\underline{x} \leq x \leq \overline{x}$ and t is capped at $[0, 0]$ for normal solves and $[0, +\infty]$ for elastic solves. The rows of this LP are just the 328 derived rows, each stored as $G_r x - t \leq h_r$, where with $t = 0$ this becomes $Gx \leq h$.

The objective is a vector with a single 1, e_v, and optimum direction. Three kinds are used

- maximize p_5
- maximize or minimize one coordinate x_v for tightening

- minimize t for elastic solves

The output is a status plus vectors depending on the status:

- optimal: x^* (with t^* as its last entry), row duals $\phi \in \mathbb{R}^{328}$, which becomes λ
- infeasible: dual ray $\phi \in \mathbb{R}^{328}$, which becomes λ
- error: no vector is returned

To convert ϕ into λ, we take the safe approach of considering both $\lambda = \max(0, \pm\phi)$, to prevent any λ with negative variables.

5.2 The Algorithm

For a box B, process_node does the following, with the end goal of pruning the node (failing to prune simply means moving to the next step):

1. Build $R(B)$ from B
2. Try to prune B directly. Ask HiGHS to maximize $c^\top x$ over $R(B)$, with $c = e_{p_5}$. There are the following possibilities
   - Infeasible: test if $Q(0, \lambda) < 0$ rigorously. If this fails, run the elastic solve and test its λ the same way. Either success means pruned.
   - Optimal: compute $Q(e_{p_5}, \lambda)$. If $Q < K$, $p_5 \le Q < K$, pruned.
   - Failed: continue to next step
3. Tighten the box for the 13 selected coordinates $p_2, p_3, p_4, u_{12}, \cdots, u_{45}$, both upper and lower bounds. Ask HiGHS to maximize $\pm x_v$ over $R(B)$ always on original B:
   - Infeasible: test it as in step 2. Success means pruned. Otherwise, move on
   - Optimal: $Q(\pm e_v, \lambda)$ is a guaranteed bound on $\pm x_v$. Widen these bounds by a small safety margin $10^{-7} + 10^{-5} \times$width and shrink the box in that dimension if that helps.
   - Failed: skip that coordinate
4. Call the shrunk box B'. Repeat step 2 but on $R(B')$, prune if possible
5. Branch by heuristically (this only affects speed, not correctness) picking a coordinate q and its (dyadic) midpoint $\mu_q = \frac{1}{2}(a_q + b_q)$. This results in two halves of B', $B'_{\le}$ and $B'_{\ge}$,

$$B' = \underbrace{B' \cap \{y_q \le \mu_q\}}_{B'_{\le}} \cup \underbrace{B' \cap \{y_q \ge \mu_q\}}_{B'_{\ge}}.$$

   that become the next children's pre-boxes. If no coordinate can be split for some reason, the node is marked unresolved.

This algorithm is implemented in Julia by an AI agent (Claude Opus 5.5) and proposes the nodes and certificates, which will then be verified in Section 6.

5.3 Soundness of the Tree

Theorem 5.1. *If the search terminates with a valid certificate for every leaf pruned and for every tightening, then* $\mathcal{F}_K = \emptyset$

Proof. For each node ν, let B_ν be its box and B'_ν its tightened Box without losing valid points. Then, we have the following properties for each node:

1. $\mathcal{F}_K \cap B_\nu \subseteq B'_\nu$. Suppose $y \in \mathcal{F}_K \cap B_\nu$. By Lemma 3.3, $\Phi(y) \in R(B_\nu)$. Then by Theorem 4.1 with $c = e_v$ (v being the dimension being shrunk) gives $y_v \leq Q(e_v, \lambda) \leq \beta$ for shrinking certificate λ, new upper bound β. So, any y must satisfy the new upper bound β. With the same argument on the lower bound, we have $y \in \mathcal{F}_K \cap B'_\nu$.

2. if ν is internal, $B'_\nu = B_{\nu_1} \cup B_{\nu_2}$;

3. if ν is a leaf, $\mathcal{F}_K \cap B'_\nu = \varnothing$. Let $y \in B'_\nu$, again by Lemma 3.3, $\Phi(y) \in R(B'_\nu)$. The node is terminated when $Q(0, \lambda) < 0$, giving direct contradiction, or when $Q(e_{p_5}, \lambda) < K$, contradicting $p_5 \geq K$ for $y \in \mathcal{F}_K$.

We wish to show that $\mathcal{F}_K \cap B_\nu = \emptyset$.

Base case: ν is a leaf: this means that $\mathcal{F}_K \cap B'_\nu = \varnothing$. Then $\mathcal{F}_K \cap B_\nu \subseteq \mathcal{F}_K \cap B'_\nu = \varnothing$. So the original box is also free of counterexamples.

Inductive step: v has two children. Then

$$\begin{aligned} \mathcal{F}_K \cap B_\nu &\subseteq \mathcal{F}_K \cap B'_\nu \\ &= \mathcal{F}_K \cap (B_{\nu_1} \cup B_{\nu_2}) \\ &= (\mathcal{F}_K \cap B_{\nu_1}) \cup (\mathcal{F}_K \cap B_{\nu_2}) \\ &= \varnothing \end{aligned}$$

This shows that for all nodes $\mathcal{F}_K \cap B_\nu = \varnothing$. At the root, $B_{root} = B_0(K) \supseteq \mathcal{F}_K$. Then, $\mathcal{F}_K = \mathcal{F}_K \cap B_{root} = \varnothing$ as desired.

□

6 The Independent Verification Program

For our verification program, we use exact rationals to ensure exact calculation. For $K = \frac{9}{2}$ for example, the input to our verification program consists of the following:

1. $K = \frac{9}{2}$ as an exact rational
2. Node log: 7543 fixed-size records each of 418 bytes. Each record holds
   (a) id, parent_id, side (0 root / 1 lower / 2 upper)
   (b) depth
   (c) code (either 0 = pruned leaf or 1 = branch)
   (d) split coordinate q and split point if code 1
   (e) the box after tightening (referred to as the "Stored Box"), as 24 pairs of doubles; notice the "Pre-box" is determined by the parent's box and its q and split point
3. Certificates: 102899 entries (99127 tightenings and 3772 leaves). Each entry gives:
   (a) its node, its kind and its objective ($c = 0$ or $c = \pm e_v$)

(b) for a tightening, the coordinate and side (v, hi/lo)

(c) the multipliers λ as sparse (row name, value) pairs, with absent rows meaning $\lambda_r = 0$

(d) the exporter's value of Q, used only as a further cross-check

We will check in two modules, the first checking that the tree is well-formed and that all the boxes of nodes are valid; the second checking that all certificates leading to pruning or shrinking are valid.

We check the following to ensure a well-formed tree:

1. Only one record, the root, has parent = -1, depth = 0, side = 0
2. All other records have proper parent_id, side either 1 or 2, depth = parent.depth + 1
3. Code-1 nodes have exactly two children with sides 1 and 2, code-0 should not have any children
4. Tree properties: every node is reachable from root; no cycles exist; we also include the check $\sum\ 2^{-\text{depth}} = 1$ adding up all leaf nodes' depths (as an additional, not absolutely necessary check).
5. Split validity: v within 1 and 24 whenever code = 1 and $\underline{B}_v < \mathit{split} < \overline{B}_v$; if equal to one side technically no problem, but will be flagged
6. Pre-box containment, ensure that $B'_\nu \subseteq B_\nu$
7. Whenever box.lo[v]>pre.lo[v] or box.hi[v]<pre.hi[v], $v \in T = \{11, 12, 13, 15, \cdots, 24\}$; also check that for every shrinking (id, v, side), the corresponding shrinking certificate exists
8. At any leaf, check that the corresponding pruning certificate exists.
9. Root box is exactly $B_0(9/2)$

For certificates, there are two types of certificates to be checked: shrinking and pruning.
For shrinking dimension q, we must have exactly

$$Q(e_q, \lambda) \le hi[q]$$
$$Q(-e_q, \lambda) \le -lo[q],$$

where the Q are based on the R(pre-box).

This ensures $\mathcal{F}_K \cap B \subseteq B'$, or Property 1 of Theorem 5.1.

For pruning a box, we use $c = 0$ in $Q(c, \lambda) < 0$, where Q is based on the stored box. The other case examined in Property 3 was not reported for $K = \frac{9}{2}$. For both certificates, we further check that $\lambda \ge 0$.

If all the mentioned checks pass, the hypotheses of Theorem 5.1 hold, which then implies $g_5 \le \frac{9}{2}$ as desired.

7 Conclusion

This proof rests on the classical results $g_3 = \frac{9}{4}$, $g_4 = 4$, Cohen's inequality and Python's exact rational arithmetic.

The same certificate-proposal program and verification can be done easily for $K = 4.3, 4.2, 4.15$ by just altering the inputs of the root box and K. Certificates for $4.5, 4.3, 4.2$ and 4.15 have been successfully generated, with 4.5 and 4.15 verified in the code provided `https:`

Table 1: Each leaf carries one pruning certificate. Tightening certificates were exported only for $K = 4.5$ and $K = 4.15$. Search times are wall-clock, with 9 workers. Verification was performed also only on $K = 4.5$ and $K = 4.15$, taking roughly 1 minute and 30 minutes, respectively. The exact rational K is the largest multiple of 2^{-20} not exceeding the target. The times were given from a 24GB memory computer.

target	exact K	nodes	leaves	tightenings	depth	t_{propose} (s)	t_{verify} (s)
4.5	$9/2$	7,543	3,772	99,127	21	26	51
4.3	$1127219/2^{18}$	31,951	15,976	—	29	132	—
4.2	$4404019/2^{20}$	97,683	48,842	—	33	426	—
4.15	$2175795/2^{19}$	225,465	112,733	3,366,236	40	1,138	1953

`//github.com/SupersonicMan12/g5-upper-bound`. Certificates, node information, generation code, verification code and more are all published in that link.

Additional information about the number of nodes and certificates is shown in Table 1.

Overall, we prove that no matrix has $p_5 \geq K$. This only translates to $g_5 \leq K$ rather than the (marginally) stronger $g_5 < K$. Showing that the set of $\mathcal{F} \cap \{p_5 \geq 4\}$ is compact can lead to the conclusion that the supremum is attained by the maximum. All inequalities can then be strengthened to

$$g_5 = \sup \rho = \max \rho < K,$$

as was standard in previous literature. However, this proof is yet to be completed for this set specifically.

Furthermore, noticing the trend of growth of the number of nodes and certificates in Table 1, it is clear that the value $K = \gamma$ itself cannot be achieved by bisection alone. Another argument would have to be found in future work to close that gap completely. On the other hand, although the same approach could apply to larger $n \geq 6$ to generate frontier bounds, the number of dimensions and rows in the linear programming grows quickly and is unsustainable.

Use of AI

This project was carried out with substantial help from AI coding agents (Claude Opus 5.5 on Devin, Cognition AI).

- **Method and code (AI):** the overall approach (the 24-parameter LU normal form, the linear relaxation of the bilinear terms, branch-and-bound with LP-duality certificates) and an initial draft of the proof were produced by AI. The AI also wrote the search and certificate-generation code in `generators/`, ran the searches (reports in `reports/`), and wrote `data_formats.md`.
- **Verification and paper (author):** I worked through the AI's proof draft until I understood it fully, then rewrote it as the paper, checking every step. I then independently wrote the exact checker `checker.py` and its specification `checklist.md`. AI was also used to explain background material, to review drafts and code, for proofreading and notation cleanup, and to check citations. All final claims and text were decided by me.

Importantly, the AI-written generators are not trusted: they only *propose* certificates. The final result rests on the mathematical argument in the paper, checked by the author, and on `checker.py`, written independently by the author, which re-verifies every certificate in exact rational arithmetic.

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